

## 4.5.4 Plant hormones (biology only)

### 4.5.4.1 Control and coordination

| Content                                                                                                                                                                                                                       | Key opportunities for skills development |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Plants produce hormones to coordinate and control growth and responses to light (phototropism) and gravity (gravitropism or geotropism). Unequal distributions of auxin cause unequal growth rates in plant roots and shoots. |                                          |
| (HT only) Gibberellins are important in initiating seed germination.                                                                                                                                                          |                                          |
| (HT only) Ethene controls cell division and ripening of fruits.                                                                                                                                                               |                                          |
| (HT only) The mechanisms of how gibberellins and ethene work are not required.                                                                                                                                                |                                          |

**Required practical activity 8:** investigate the effect of light or gravity on the growth of newly germinated seedlings.

Record results as both length measurements and as careful, labelled biological drawings to show the effects.

AT skills covered by this practical activity: AT 1, 3, 4 and 7.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#).

### 4.5.4.2 Use of plant hormones (HT only)

| Content                                                                                                                                                           | Key opportunities for skills development                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Students should be able to describe the effects of some plant hormones and the different ways people use them to control plant growth.                            | WS 1.3, 1.4                                                                                |
| Plant growth hormones are used in agriculture and horticulture.                                                                                                   | Understand how the everyday use of hormones as weed killers has an effect on biodiversity. |
| Auxins are used: <ul style="list-style-type: none"><li>• as weed killers</li><li>• as rooting powders</li><li>• for promoting growth in tissue culture.</li></ul> |                                                                                            |
| Ethene is used in the food industry to control ripening of fruit during storage and transport.                                                                    |                                                                                            |
| Gibberellins can be used to: <ul style="list-style-type: none"><li>• end seed dormancy</li><li>• promote flowering</li><li>• increase fruit size.</li></ul>       |                                                                                            |